A Dheeraj

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Professional Summary

Backend / Applied AI Engineer with strong Python experience and hands-on work in ML-driven systems, APIs, and data pipelines. Built real-time ML applications, backend services, and analytics systems. Interested in scalable, low-hallucination AI agents and production-grade cloud systems.

Education

B.E in Mechanical Engineering — 2022 - 2026
RV College of Engineering — *Current CGPA: 8.41*

Experience

Intern | Hindustan Aeronautics Limited (HAL) – LCA Tejas Division — Aug 2025 – Oct 2025

- Worked in a high-reliability engineering environment, emphasizing correctness, validation, and system safety.
- Gained exposure to large-scale system integration, documentation, and disciplined engineering workflows.

Projects

Real-Time Driver Monitoring System | Link | *Mediapipe, YOLOv8, Streamlit, MongoDB* — Oct 24 - Feb 25

- Built a real-time ML inference pipeline using Python, MediaPipe, and YOLOv8 for continuous driver state monitoring.
- Designed a backend service to process video streams and trigger rule-based alerts with low false positives.
- Deployed a Streamlit-based interface backed by MongoDB for structured event storage and retrieval. Achieved 90% accuracy, focusing on reliability over hallucination in safety-critical conditions.

Stock Market Management | Link | *ReactJS, Node.js, ExpressJS, Chart.js, Puppeteer, PostgreSQL* — Aug 24 - Nov 24

- Engineered a full-stack financial platform using the MERN stack (React.js, Node.js, ExpressJS) and PostgreSQL to simulate stock market trading.
- Implemented features for user/admin management, real-time stock tracking, and financial news aggregation using Puppeteer for web scraping automation.
- Designed and implemented RESTful backend APIs using Node.js and PostgreSQL for user portfolios, transactions, and analytics. Built data ingestion pipelines using Puppeteer for automated financial data collection. Integrated an LLM-powered query agent (LLaMA-3 70B) that converted natural language questions into backend database queries. Focused on controlled query generation to avoid unsafe or hallucinated outputs.

Banking Risk Analysis & Dashboarding | *Excel, Python, SQL, Power BI* — Jun 24 - Aug 24

- Designed an end to end analytics pipeline, acquired raw banking data via Excel, cleaned and transformed using Python, and structured into a MySQL database; enabled seamless querying with SQL joins/aggregations to prepare data for analysis.
- Performed Exploratory Data Analysis (EDA) in Python, uncovered insights such as 66% of customers holding a single credit card and strong positive correlation among deposits in savings, checking, and foreign currency accounts.
- Built an interactive Power BI dashboard with 10+ KPIs (Total Loan, Total Deposit, Credit Card Balance, Engagement Days, Processing Fees); enabled stakeholders to analyze loan risk profiles, client segmentation by income bands ($\leq$ 100k = Low, $\leq$ 300k = Mid), and lending trends across demographics.

Publications

Extraction of BioFuel from Algae — July 25

- Co-authored a research paper exploring innovative method for extracting biofuel from algae, highlighting sustainable energy solutions and advancing knowledge in bioenergy applications at the International Conference on Digital Technologies for Business Excellence and Sustainable Development, IIT (ISM) Dhanbad.

Certification & Achievements

JPMorgan Chase & Co. Software Engineering Job Simulation | *Forage*

- Completed a multi-task simulation to build ”Midas Core,” a financial system using Java 17 and Spring Boot.
- Demonstrated practical skills by integrating Apache Kafka for asynchronous transaction processing, using SQL & JPA with an H2 database for data persistence and validation, and building/consuming REST APIs for microservice integration.

Amazon ML Summer School | *Amazon Science India* July 25 - Aug 25

- Completed a selective, multi-week program on core and advanced machine learning topics, including Deep Neural Networks, Generative AI, and Causal Inferencing.

Skills

Programming Languages: Java, Python, SQL

Backend & Systems: Python (APIs, data pipelines, ML inference), REST APIs, backend service design, Node.js, PostgreSQL, MongoDB, Event-driven processing, real-time systems

Machine Learning & AI: Computer Vision (YOLOv8, MediaPipe), LLM integration (LLaMA-3), Model evaluation & reliability, Rule-based + ML hybrid systems

Cloud & DevOps (Learning / Basic) : Docker (containerization – learning), AWS / GCP fundamentals (compute, storage – learning), Scalable system design concepts

Currently Exploring

- Python-based AI agents with guardrails and constrained generation
- Prompt engineering for low-hallucination and controlled outputs
- Dockerized backend services for ML and API-based systems
- Cloud deployment fundamentals using AWS and Google Cloud Platform